Assignment 02

Comparing Predictive Models Honestly

Teaching Week 08 | Individual work. Submit in R or Python.

Purpose

This assignment joins the predictive work of Weeks 06 to 09 into one study, in place of Labs 06 to 09. On one data set you will compare a baseline, k-nearest neighbours, linear discriminant analysis, a support vector machine and a decision tree. Every model is tuned inside cross-validation and scored outside it, on the same folds, and Class 13's corrected test decides whether any of them is genuinely better. The marks fall on the protocol and the argument, not on the highest score.

Learning Outcomes

After completing this assignment, you should be able to

- define a classification target, the unit of prediction, and the time order and natural groups of the rows;
- prepare predictors so that every learned step is fitted on training rows only;
- tune kNN, a support vector machine and a decision tree inside cross-validation and score each outside it;
- compare several models on the same folds with a class-aware metric and Class 13's corrected resampled t-test;
- explain where a model fails, using the confusion pattern and the rules of a fitted tree; and
- publish a reproducible analysis that another analyst can rerun from the submitted materials.

The Dataset

Choose one of the two records below and keep it for the whole assignment. Both come from the Week 08 lab catalogue, and both record when and where each row was collected, which matters for an honest estimate.

Option A | Beaufort Sea Fish Measurements

Publisher. Fisheries and Oceans Canada
Official record. https://open.canada.ca/data/en/dataset/74e3dbd2-1845-11ef-90aa-8b219c568296
Record identifier. `74e3dbd2-1845-11ef-90aa-8b219c568296`
Licence and format. Open Government Licence – Canada; CSV, with a PDF data dictionary
Expected variables. station; latitude; longitude; date; gear; family; species; weight; total, fork and standard length; liver and gonad weight; sex; maturity
Use in this assignment. classify a fish's family from its measurements and where it was caught
Why it fits. About 10 000 fish from 75 trawl stations sampled between 10 August and 9 September 2013, in 14 families. Two thirds are cod (Gadidae), so group the rare families into one class with a stated rule and use a class-aware metric. Liver weight is missing for almost every fish, and gonad weight, sex and maturity for more than half, so decide what each gap means before using the column. Fish from one station share conditions, so the station is a natural group.

Option B | FunRoots Tree-Seedling Traits

Publisher. Natural Resources Canada
Official record. https://open.canada.ca/data/en/dataset/d9c4d93f-2388-4158-b3cf-49a9ecea6b2c
Record identifier. `d9c4d93f-2388-4158-b3cf-49a9ecea6b2c`
Licence and format. Open Government Licence – Canada; CSV tables, with a data dictionary and a protocol
Expected variables. seedling; region; light condition; plot; species; sampling date; age; height; stem diameter; crown; leaf and root traits; biomass
Use in this assignment. classify a seedling's species, or its light condition, from its traits
Why it fits. The seedlings table holds 338 seedlings of seven species, 20 to 61 each, from 14 plots in three regions, under open or closed canopy, with dozens of size, leaf and root measurements. Several traits are derived from others, so justify every predictor you keep. Seedlings from one plot share their site, so the plot is a natural group.

Download the files yourself, read the publisher's dictionary before writing any analysis code, and cite the record. Record the access date and the size of every file you read.

Data Contract

Your modelling table must satisfy every rule below before you fit anything.

- Define one classification target with at least three classes, and keep at least 300 complete rows for option A or 250 for option B.
- Keep at least four numeric predictors with genuine variation, and justify each one from the dictionary. Identifiers and the target's own codes are not predictors.
- State the rule for every class you merge, every row you drop, and every gap you fill, and report the row and class counts before and after.

- Keep the date and the station or plot of every row, for the evaluation design and the limitations, even when they are not predictors.
- Record the official URL, publisher, licence, retrieval steps, access date and file sizes.

What the Contract Does Not Settle

Meeting the contract shows that your table is faithful to the published record. It does not show that your folds are honest, that your comparison is fair, or that the best model would stay the best on another season of data.

Modelling Protocol

One Set of Folds for Every Model

Cut the rows into five stratified outer folds, and inside each outer training set cut five stratified inner folds, as Class 14 did. Freeze the folds once, in a file or an object your code reads, and use them for every model, so every comparison is paired. Choose every setting on the inner folds and score each model once on the outer fold it never saw. Put scaling and any imputation inside the pipeline, so they are refitted on every training set.

Five Models

Fit a baseline that always predicts the most common class; linear discriminant analysis; kNN with K chosen inside; a radial support vector machine with C and γ chosen inside on a grid of powers of ten; and a decision tree with its depth or pruning level chosen inside. Report the grid you searched for each model. If a chosen setting sits on the edge of its grid, widen the grid.

A Class-Aware Metric

Report balanced accuracy, the mean recall over the classes, as the headline metric for every model, together with the confusion matrix of the model you recommend. Plain accuracy may be reported beside it, never instead of it.

Reproducible Workflow

You may complete and submit the assignment in either R or Python. Both paths are marked against the same evidence and the same criteria, and both must run end to end from one documented entry point in your portfolio repository.

Use the methods taught in the course. Write your own preparation, fitting, evaluation and reporting source, and generate every reported number, table and figure from it. Record the software, package versions, random seeds and the exact command that reproduces the analysis.

Work Programme

1. Register the Source and Ask the Question

Record the provenance the contract requires. State the target, the unit of prediction, and the decision a working model would support, in one sentence each.

2. Build the Modelling Table

Build the table the contract describes. Publish a variable table giving each kept column its type, its role, and its share of missing values. Report the class counts of the target.

3. Fix the Folds

Create the outer and inner folds, report the class counts in each outer fold, and show that the same folds reach every model.

4. The Week 06 Models

Fit the baseline, LDA and kNN. Report the inner scores across K and the outer scores of the chosen K. Explain why kNN needs scaled predictors and LDA does not.

5. Is the Estimate Honest?

Compare the outer score of your tuned kNN with the best inner score its search reported, and explain the gap with Class 13's lucky-search argument. Then rescore kNN by training on earlier rows and testing on later ones, with Class 13's time-ordered windows, and explain how and why the score changes. State which estimate answers the question a user of your model would ask, and what the stations or plots add to that limitation.

6. The Support Vector Machine

Tune C and γ inside. Show the inner grid for one outer fold and explain what its rows, columns and diagonals mean. If the classes are very unequal, try class weights as one more setting and judge them with balanced accuracy.

7. The Decision Tree

Tune the depth or the pruning level inside. Plot training and outer balanced accuracy against tree size, and read the rules of one fitted tree in the original units. Name one rule that matches domain knowledge and one that looks like an accident of these rows.

8. Compare and Conclude

Tabulate the outer score of every model on every fold, with the mean. Test the gap between your best model and its closest rival with Class 13's corrected resampled t-test, stating the overlap term you used. Recommend one model, show its confusion matrix, describe which classes it confuses and why, and state what the evidence does not show.

What to Submit and Where

Keep the work under `assignments/assignment-02/` in a private work repository with instructor access until the deadline has passed and the instructor opens sharing. Commit the finished report, source, and evidence first. In a second commit, record that completed-work commit's full SHA and immutable commit URL in the directory's `README.md`. Push both commits to the default branch by the deadline. A commit cannot contain its own SHA; the marker assesses the snapshot named in the pushed README. The assignment is released on October 27 and is due at 23:59 on November 16, in the course time zone.

Include

- a concise PDF report carrying the question, the variable table, the protocol, the results for every model, the comparison and test, and the limitations;
- the complete R or Python source, plus every table, metric and figure the report cites, written by that source rather than edited by hand;
- the frozen fold file, or the code that creates it together with its seed;
- a README carrying the submitted commit SHA and its commit URL, the official data URL, the retrieval steps, the access date and file sizes, the exact execution command, and the runtime environment; and
- no redistributed copy of the source data, which is linked rather than committed.

A submission is complete only when the required files open and the analysis can be reproduced from the submitted materials. Late work loses 5 percentage points of the available assignment mark for each late calendar day, for up to seven calendar days, subject to the syllabus's accommodation provisions. Cite the dataset and every non-trivial external source, and submit only your own work.

When sharing opens, publish the submitted commit with its existing history for discussion at the next class meeting, if one follows, and in your portfolio. Make corrections in new commits. Do not amend, rebase, squash, force-push, delete, or backdate submitted history. The pushed README record establishes what was submitted and when; Git timestamps alone do not prove timely completion.

Generative AI output is not authorized for this assignment; the syllabus's academic integrity and private-repository exception provisions apply.

Grading

This assignment is worth 7.5 percent of the course grade. The weights below are percentage points of that course grade and sum to the assignment's weight.

Table 1. Assignment 02 marking criteria.

Criterion	Course weight (%)
Provenance, target and modelling table. Record documented, target and predictors justified, merges and gaps stated with counts.	1.0
Fold design and leakage control. Frozen folds shared by every model, every learned step fitted inside the training rows.	1.5
Week 06 models and the honesty checks. Baseline, LDA and tuned kNN reported; lucky-search gap and time-ordered rescoring explained.	1.5
SVM and tree. Both tuned inside with the grid reported, SVM grid read, tree curve and rules interpreted.	1.5
Comparison and conclusion. Paired fold table, corrected test with its overlap term, confusion pattern explained, limits stated.	1.5
Reproducibility and communication. Runs end to end from one entry point, artefacts generated, report clear and cited.	0.5
Total	7.5

How This Relates to the Weekly Labs

This assignment replaces Labs 06 to 09. Their methods, kNN and LDA from Week 06, honest performance estimation from Weeks 07 and 08, support vector machines from Week 08 and trees from Week 09, are assessed here once, on one data set, so that the comparison between them is fair. The other weekly labs are unchanged, and the best seven of the eight count toward the lab component.